Functional Profiling of Gut Microbiome of Children in Atlanta, USA and Maputo, Mozambique Using HUMAnN3

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Practical Introduction to Metagenomics

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May 11, 2026

Abstract

Shotgun metagenomic sequencing is a useful tool for studying microbial communities because it allows for characterization without the need for culturing individual organisms. In this project, publicly available stool metagenomes were analyzed to compare the functional potential of gut microbiomes from children in Atlanta, USA and Maputo, Mozambique. Three samples from each location were obtained from the NCBI Sequence Read Archive and processed using a workflow that included quality filtering, taxonomic profiling with MetaPhlAn 4, and functional profiling with HUMAnN 3. Pathway abundance tables were combined, normalized to relative abundance, and analyzed in Python. Metabolic pathways involved in amino acid biosynthesis, nucleotide biosynthesis, and metabolism were identified across all six samples. Many pathways, including queuosine biosynthesis, guanosine ribonucleotide biosynthesis, and L-methionine biosynthesis, were more abundant in the Maputo samples, while glycogen biosynthesis was relatively more abundant in the Atlanta samples. Although no pathways reached statistical significance due to the small sample size, the results suggest that differences in geography, diet, and environmental exposures could influence the functionality of the gut microbiome.

Introduction

Metagenomics is the direct study of genetic material from environmental samples. Traditional genomics relies on isolating and culturing individual organisms in order to sequence the DNA. However, by employing metagenomics, all DNA present in a sample can be sequenced using shotgun metagenomic sequencing without the need for culture or isolation. Metagenomics makes it possible to study complex environments while getting a dynamic picture of all the microorganisms in a sample. Shotgun metagenomics can also identify metabolic pathways that are involved in metabolism, antibiotic resistance, virulence, and other biological processes (Quince et al., 2017). Due to these attributes, shotgun metagenomics can be used as a valuable tool for microbiome research.

The human gut microbiome plays a central role in digestion, immune regulation, and metabolism. Differences in geography, diet, and environment can have a strong influence on not only the taxonomic make up of gut microbial communities, but also functional capacity (Turnbaugh et al., 2007). By using shotgun metagenomic sequencing these gut communities can be studied to profile and compare microbial taxa and the genes and pathways they encode. Taxonomic differences between different populations are widely documented, but performing a functional comparison between populations can provide further insight into the metabolic differences of microbial communities (Turnbaugh et al., 2007).

Studies have shown that children living in low- and middle-income countries have higher overall gut microbiome diversity and have an abundance of fiber-degrading organisms and increased metabolic pathways that are involved in carbohydrate utilization and biosynthesis (Yatsunenko et al., 2012). The presence of organisms like *Prevotella* and *Faecalibacterium* are typically higher in low/middle income countries. In contrast, children from high income countries have gut microbiomes that have less diversity and are associated with organisms like *Bacteroides* and *Bifidobacterium* (Wu et al., 2011). These differences have the potential to influence nutrient extraction, immune development, and susceptibility to disease.

HUMAnN (The HMP Unified Metabolic Analysis Network) is a tool that maps metagenomic reads to gene families and reconstructs pathway abundances (Franzosa et al., 2018). This tool allows for a direct comparison of metabolic potential from different microbial communities. The goal of this project was to compare the functional potential of children (ages 2 and under) gut microbiomes from Atlanta, USA and Maputo, Mozambique using publicly available shotgun metagenomic data. Based on previous studies of gut microbiomes, pathways involved in amino acid, nucleotide, and carbohydrate metabolism were expected to be more abundant in the Maputo samples (Yatsunenko et al., 2012).

Methods and Materials

Stool metagenomes were obtained from the NCBI Sequence Read Archive (SRA) from the BioProject PRJNA747761. Three samples from children in Atlanta, USA (SRR15202440, SRR15202441, and SRR15202442) and three samples from children in Maputo, Mozambique (SRR15209203, SRR15209176, and SRR15209171) were selected. Raw sequencing reads were quality filtered using fastp and paired-end reads were merged into FASTQ files for analysis (Chen et al., 2018). Taxonomic profiling was performed using MetaPhlAn 4 with the mpa_vJun23_CHOCOPhlAnSGB_202307 database (Beghini et al., 2021). Functional profiling was performed using HUMAnN 3. Pathway abundance tables from all six samples were combined and normalized to relative abundance. Analysis was performed with Python using pandas, matplotlib, seaborn, and SciPy. The 20 most abundant pathways were identified and compared between Atlanta and Maputo samples. Mann–Whitney U tests were used to assess pathway-level differences between groups.

Results

Functional Profiling

Using HUMAnN, pathway abundance profiles were generated for all six samples. After combining and normalizing the pathway abundance tables, relative abundances were compared across samples. Pathways involved in central metabolism, amino acid biosynthesis, nucleotide biosynthesis, and cell wall synthesis were most prevalent across all samples, indicating a strong shared functional importance within the gut microbiomes (Figure 1). The most abundant pathways across all samples included L-rhamnose biosynthesis, L-valine biosynthesis, glycolysis IV, guanosine ribonucleotide biosynthesis, and L-isoleucine biosynthesis. These pathways are essential for energy production, protein synthesis, nucleotide synthesis, and bacterial cell wall formation.

Figure 1: Overall Top 20 Most Abundant Metabolic Pathways

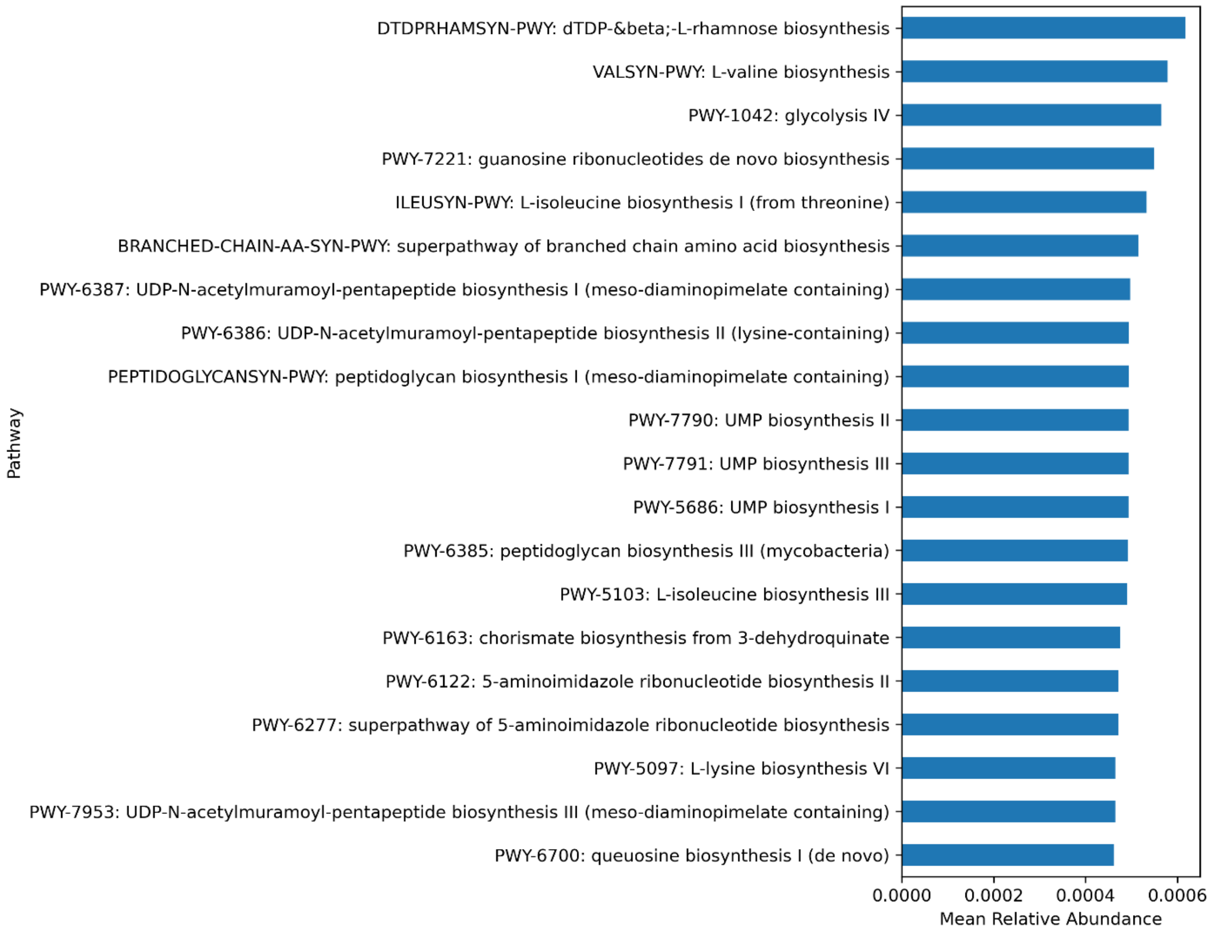

Figure 1: Top 20 most abundant metabolic pathways across all six gut metagenomes. The values represent the mean relative abundance across the Atlanta and Maputo samples.

The heatmap of the top 20 pathways generally showed similar functional profiles across all samples, with the exception of SRR15209203 from the Maputo cohort which showed higher abundance overall in most profiles (Figure 2).

Figure 2: Top 20 Most Abundant Metabolic Pathways Heatmap

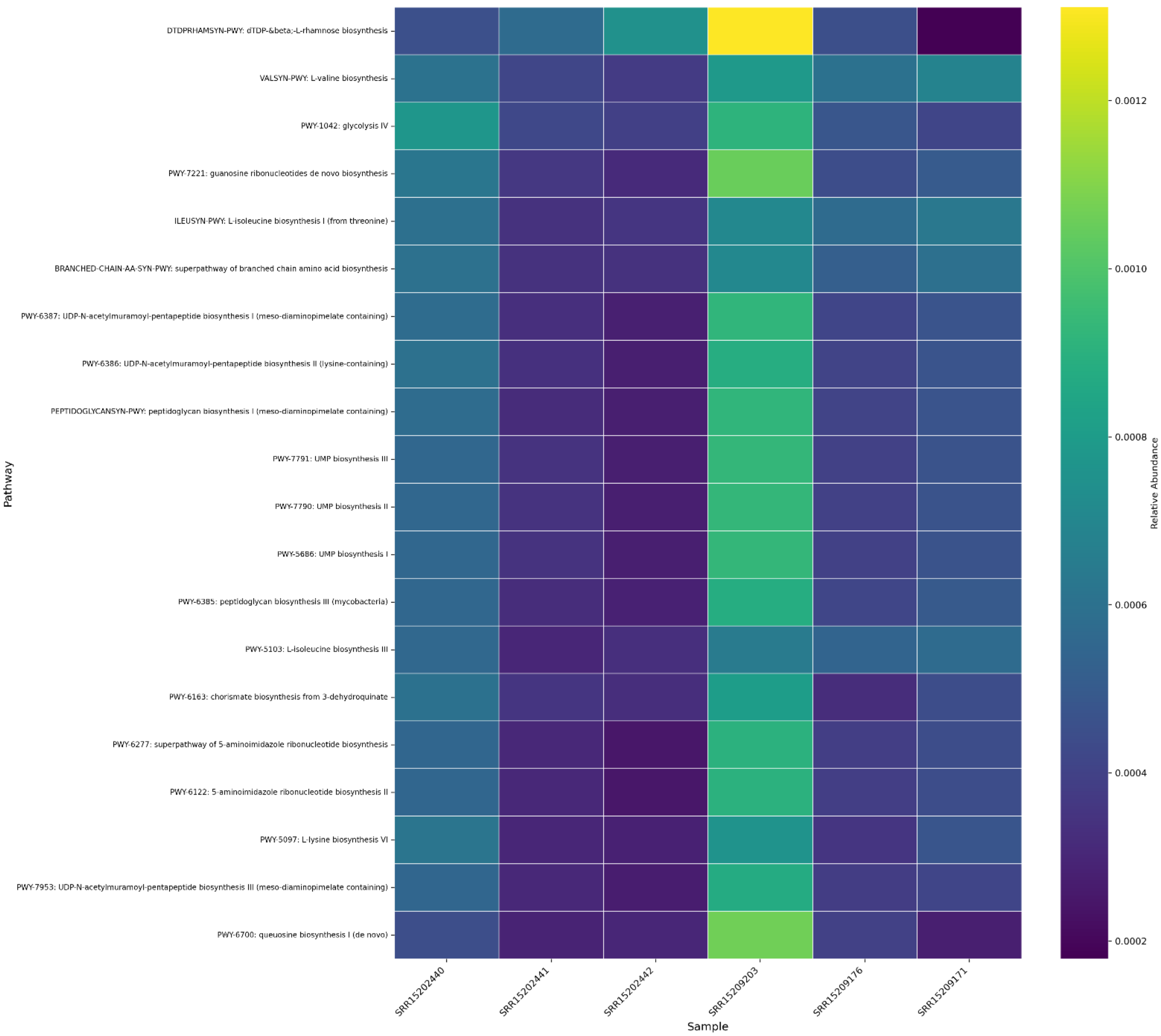

Figure 2. Heatmap of the 20 most abundant metabolic pathways across gut microbiome samples. Each column represents one sample: Atlanta, USA (SRR15202440, SRR15202441, and SRR15202442) and Maputo, Mozambique (SRR15209203, SRR15209176, and SRR15209171).

Comparing Maputo Mean Abundance to Atlanta Mean Abundance

In comparing the top pathway abundances from the two groups, almost all pathways were more abundant in the Maputo samples (Figure 3). Queuosine biosynthesis, guanosine ribonucleotides de novo biosynthesis, L-methionine biosynthesis, and branched-chain amino acid biosynthesis are enhanced in the Maputo samples. These pathways are involved in nucleotide production, amino acid synthesis, and bacterial cell wall formation, which suggests increased biosynthetic potential in the Maputo microbiomes (Wu et al., 2011; Yatsunenko et al., 2012). In contrast, glycogen biosynthesis, which was relatively more abundant in the Atlanta samples, suggests differences in carbohydrate storage and energy metabolism between the two microbial communities.

Figure 3: Top Pathway Differences (Maputo Mean Abundance−Atlanta Mean Abundance)

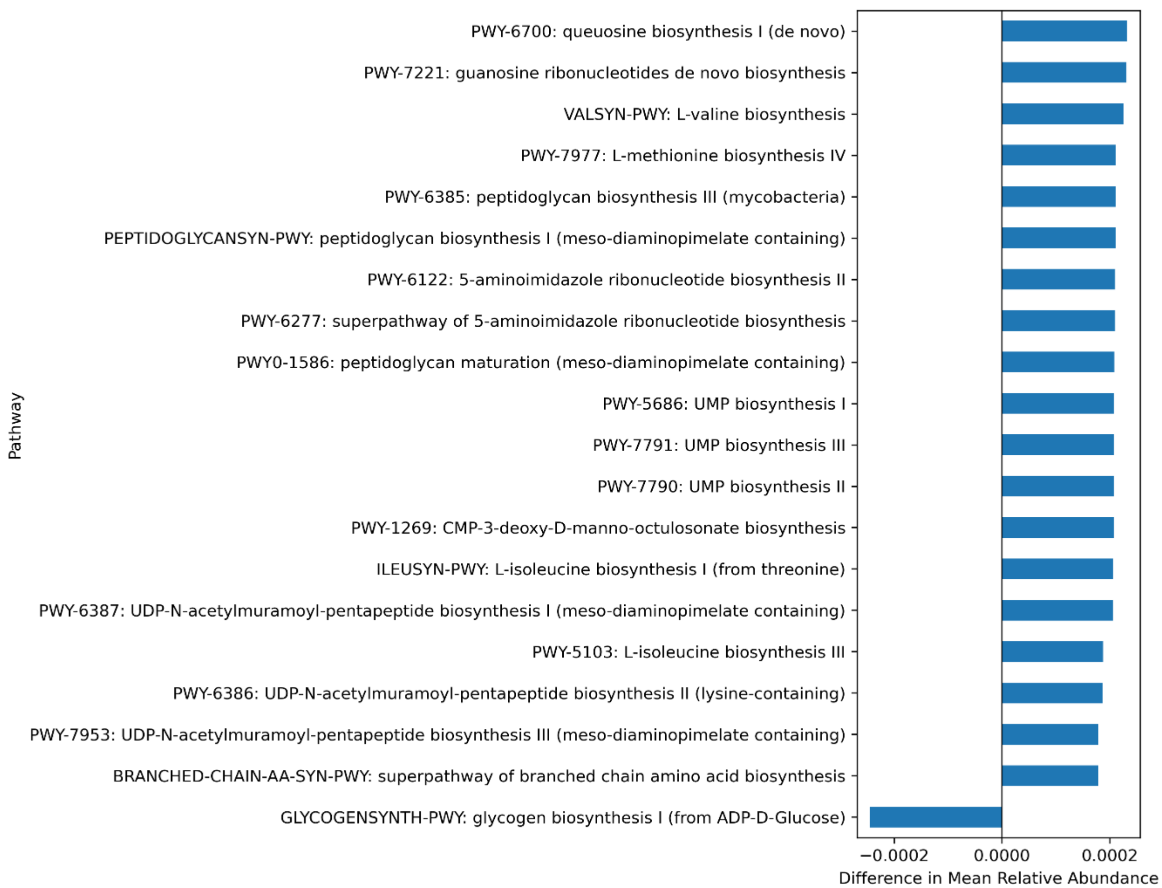

Figure 3. Top 20 metabolic pathways showing the differences in mean relative abundance between gut microbiomes from Maputo, Mozambique and Atlanta, USA. Positive abundance indicates that pathways are enriched in the Maputo samples, while negative abundance indicates that pathways are enriched in the Atlanta samples.

Statical Analysis

Mann–Whitney U tests were used to compare pathway abundances between the Atlanta and Maputo groups. None of the pathways met standard thresholds for statistical significance, which is to be expected due to the small sample size.

Discussion

The functional differences between the Atlanta and Maputo gut microbiomes reflect differences in microbial community composition, diet, and environmental exposures. Initial MetaPhlAn profiling of the Maputo samples showed an abundance of *Prevotella* species, which is associated with high-fiber diets typical of non-Western populations (Wu et al., 2011; Yatsunenko et al., 2012). Gut microbial communities with an abundance of *Prevotella* are often enriched for pathways involved in amino acid biosynthesis, nucleotide synthesis, and carbohydrate metabolism, which is consistent with the results (Wu et al., 2011; Yatsunenko et al., 2012). The enrichment of queuosine biosynthesis in the Maputo samples is noteworthy because queuosine is a modified nucleoside that is involved in translational accuracy and bacterial stress responses (Nayfach et al., 2021). Also, increased abundance of guanosine ribonucleotides and L-methionine IV biosynthesis pathways suggests a greater amount of biosynthetic activity in these microbiomes compared to the Atlanta samples. In contrast, the higher abundance of glycogen biosynthesis in the Atlanta samples could indicate differences in carbon storage between dominant taxa. Although statistical significance was limited, the observed trends support the hypothesis that gut microbiomes from distinct geographic regions exhibit functional differences.

This study has several limitations. The sample size was very small, which limited the statistical significance of the results and affected the ability to detect significant differences. HUMAnN assumes functional potential based on gene content and does not measure actual gene expression or metabolic activity. Lastly, sequencing depth varies across samples, which can influence pathway estimates even with normalization.

Conclusion

This project demonstrates a complete shotgun metagenomics workflow for comparing the functional potential of gut microbiomes from children in Atlanta, USA and Maputo, Mozambique. Using publicly available stool metagenomes, metabolic pathways present across all six samples were identified, with several pathways involved in amino acid and nucleotide biosynthesis being more abundant in the Maputo samples. These findings suggest that differences in geography, diet, and environmental exposures may influence the metabolic capabilities of the gut microbiome.

References

Quince, C., Walker, A. W., Simpson, J. T., Loman, N. J., & Segata, N. (2017). Shotgun metagenomics, from sampling to analysis. Nature Biotechnology, 35, 833-844.

Yatsunenko, T., Rey, F. E., Manary, M. J., et al. (2012). Human gut microbiome viewed across age and geography. Nature, 486, 222-227.

Beghini, F., McIver, L. J., Blanco-Míguez, A., et al. (2021). Integrating taxonomic, functional, and strain-level profiling of diverse microbial communities with bioBakery 3. eLife, 10, e65088.

Bolger, A. M., Lohse, M., & Usadel, B. (2014). Trimmomatic: A flexible trimmer for Illumina sequence data. Bioinformatics, 30(15), 2114–2120.

Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). fastp: An ultra-fast all-in-one FASTQ preprocessor. Bioinformatics, 34(17), i884–i890.

Franzosa, E. A., McIver, L. J., Rahnavard, G., et al. (2018). Species-level functional profiling of metagenomes and metatranscriptomes. Nature Methods, 15, 962–968.

Nayfach, S., Roux, S., Seshadri, R., et al. (2021). A genomic catalog of Earth's microbiomes. Nature Biotechnology, 39, 499–509.

Segata, N., Waldron, L., Ballarini, A., et al. (2012). Metagenomic microbial community profiling using unique clade-specific marker genes. Nature Methods, 9, 811–814.

Turnbaugh, P. J., Ley, R. E., Hamady, M., et al. (2007). The human microbiome project. Nature, 449, 804–810.

Wu, G. D., Chen, J., Hoffmann, C., et al. (2011). Linking long-term dietary patterns with gut microbial enterotypes. Science, 334(6052), 105–108.

Appendix

See the NSherman_Final_USER_GUIDE.txt